Engineering Specification Document

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SP-X

Generic MegaRAC® SP-X Firmware AMI YAFU Commands Specification

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		& Jabastin Doss A	

Contents

1 YAFU Commands

This section lists the YAFU related commands.

***Note**: YAFU firmware update commands need to be used in a sequence and are not meant to be used individually. YAFU commands are for firmware upgrade, should be used with more care, otherwise it can impact the exist firmware running in the BMC.*

1.1 YAFU_Get_Status Command

This command is used to do get FW status to make sure the device is responding.

NetFn	0x32
Command	0x04

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0004 (LSB first)
7:8	Datalen. Fixed 0x0000
9:12	CRC32chksum.(byte values will be ignored)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0004 (LSB first)
8:9	Datalen. Fixed 0x0008 (LSB first)
10:13	CRC32chksum. Checksum of Byte14~21(because Datalen=8)
14:15	LastStatusCode YAFU_CC_NORMAL 0x00

	YAFU_CC_INVALID_DATLEN 0xcc YAFU_CC_ALLOC_ERR 0x02 YAFU_CC_DEV_OPEN_ERR 0x03 YAFU_CC_SEEK_ERR 0x04 YAFU_CC_READ_ERR 0x05 YAFU_CC_WRITE_ERR 0x06 YAFU_CC_MEM_ERASE_ERR 0x07 YAFU_CC_IN_DEACTIVATE 0x08 YAFU_OFFSET_NOT_IN_ERASE_BOUNDARY 0x09 YAFU_SIZE_NOT_IN_ERASE_BOUNDARY 0x10 YAFU_FLASH_ERASE_FAILURE 0x11 YAFU_INVALID_CHKSUM 0x12 YAFU_ERR_STATE 0x13 YAFU_ECF_PROGRESS 0x14 YAFU_ECF_SUCCESS 0x15 YAFU_VERIFY_SUCCESS 0x16 YAFU_VERIFY_PROGRESS 0x17 YAFU_CC_GET_MEM_ERR 0x18 YAFU_CC_INVALID_DATA 0x19 YAFU_CC_INVALID_SIGN_IMAGE 0x20 YAFU_CC_SIGNED_SUPP_NOT_ENABLED 0x21 YAFU_VERIFY_FAILURE 0x22 YAFU_NOT_SUPPORTED 0x23 YAFU_FLASHING_SAME_IMAGB 0x24
16:17	YAFUState. Fixed 0x0000=Success
18:19	Mode. Fixed 0x0000
20:21	Reserved. Fixed 0x0000
22:86	Message

1.2 YAFU_Switch_Flash_Device Command

This command is used to switch flash device.

NetFn	0x32
Command	0x52

Request Data:

Byte	Data field
1	SPI Device BMC 0x00 BOIS 0x01 CPLD 0x02 BMV_MMC 0x03

Response Data:

Byte	Data field
1	Completion Code
2:32	MTDName represented in ASCII format
33:36	Flash Size. Size of flash memory(LSB first)
37:40	Flash Address. Starting address of the flash memory (LSB first)
41:44	Flash Erase Block Size. (LSB first)
45:46	Flash Product ID. (LSB first)
47	Flash Width
48	FMH Compliance
49:50	Reserved. Fixed 0x0000
51:52	Number Of Erase Blocks (N)

1.3 YAFU_Get_Flash_Info Command

This command is used to get the basic information about the Flash Hardware.

NetFn	0x32
Command	0x01

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0001 (LSB first)
7:8	Datalen. Fixed 0x0000
9:12	CRC32chksum. (byte values will be ignored)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0001 (LSB first)
8:9	Datalen. =32 or 32+N
10:13	CRC32chks-um. Checksum of Byte14~33
14:17	Flash Size. Size of flash memory (LSB first)
18:21	Flash Address. Starting address of flash memory (LSB first)
22:25	Flash Erase Block Size (LSB first)
26:27	Flash Product ID. Fixed 0x0000
28	Flash Width. Fixed 0x08
29	FMH Compliance. Fixed 0x01
30:31	Reserved. Fixed 0x0000

32:33	Number Of Erase Blocks (N). (LSB first)

***Note**: If Number of Erase Blocks * Erase Block Size == Total Flash Size, then data length = 32, else data length = 32 + Number of Erase Blocks.*

1.4 YAFU_Get_FMH_Info Command

This command is used to get the basic information about the FMH.

NetFn	0x32
Command	0x03

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0003 (LSB first)
7:8	Datalen. Fixed 0x0000
9:12	CRC32chksum. (byte values will be ignored)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0003 (LSB first)
8:9	Datalen. =4+N*64 where N is the number of FMH sections returned in data [16:17] (LSB first)
10:13	CRC32chksum. Checksum of Byte14~18+N*64
14:15	Reserved. Fixed 0x0000
16:17	Number of FMH (N). (LSB first)
18:18+N*64	FMH Information

1.5 YAFU_Erase_Copy_Flash Command

This command is used to erase the required sectors and then copy the contents from memory to Flash. On Success the Size Copied equals to Size to copy.

NetFn	0x32
Command	0x26

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0026 (LSB first)
7:8	Datalen. Fixed 0x000c (LSB first)
9:12	CRC32chksum. Checksum of Byte13~24
13:16	Source Memory offset. Offset location from where the contents are to be copied from. (LSB first)
17:20	Destination Flash offset. Flash memory offset where the contents are to be written. (LSB first)
21:24	Size to copy. Number of bytes to copy from source memory offset to the destination flash offset (LSB first)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0026 (LSB first)
8:9	Datalen. Fixed 0x0004 (LSB first)
10:13	CRC32chksum. Checksum of Byte14~17
14:17	Number of bytes Copied. (LSB first)

1.6 YAFU_Get_Ecf_Status Command

This command is used to get upgrade status to check ECF progress completed.

NetFn	0x32
Command	0x28

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0028 (LSB first)
7:8	Datalen. Fixed 0x0000
9:12	CRC32chksum. (byte values will be ignored)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0028 (LSB first)
8:9	Datalen. Fixed 0x0000
10:13	CRC32chksum
14:17	Status. (LSB first) Failure 0x0000 Success 0x0001
18:19	Progress (% completed). LSB first

1.7 YAFU_Activate_Flash_Mode Command

This command is used to inform device to enter into Flash mode.

NetFn	0x32
Command	0x10

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0010 (LSB first)
7:8	Datalen. Fixed 0x0002 (LSB first)
9:12	CRC32chksum. Checksum of Byte13~14
13:14	Mode Type. LSB first 0x0000 - Normal mode 0x0001 - Recovery mode
15	Migration 0x00 - kernel-3 code base version 0xf5 - Kernel-5 code base version This data field is used to check whether the image to be flashed is from kernel 3 to kernel 5.

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0010 (LSB first)
8:9	Datalen. Fixed 0x0002 (LSB first)
10:13	CRC32chksum
14	Delay (in Seconds). Number of seconds needed for the device to activate flash mode

1.8 YAFU_Read_Flash Command

This command is used to read Flash Contents.

NetFn	0x32
Command	0x22

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0022 (LSB first)
7:8	Datalen. Fixed 0x0007 (LSB first)
9:12	CRC32chksum. Checksum of Byte13~21
13:16	Offset to Read. Flash memory offset address from where contents are to be read (LSB first)
17	Read Width
18:21	Size to Read. Number of bytes to read (LSB first)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0022 (LSB first)
8:9	Datalen. = Size to Read (LSB first)
10:13	CRC32chksum. Checksum of Byte14~14+N
14:14+N	Flash Contents Read, where N is the datalen

1.9 YAFU_Get_Boot_Vars Command

This command is used to get all the Boot variable values.

NetFn	0x32
Command	0x42

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0042 (LSB first)
7:8	Datalen. Fixed 0x0000
9:12	CRC32chksum. (byte values will be ignored)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum.
6:7	YafuCmd. Fixed 0x0042 (LSB first)
8:9	Datalen. = N+1 (LSB first)
10:13	CRC32chksum. Checksum of Byte14~15+N
14	Variable Count(N), depends on the boot variables
15:15+N	ASCIIZ Variable Name

1.10 YAFU_Get_Boot_Config Command

This command is used to get the necessary Boot configuration.

NetFn	0x32
Command	0x40

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0040 (LSB first)
7:8	Datalen. Fixed 0x0041 (LSB first)
9:12	CRC32chksum. Checksum of Byte13~77
13:77	ASCIIZ Variable Name.

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0040 (LSB first)
8:9	Datalen. = N+1 (LSB first)
10:13	CRC32chksum. Checksum of Byte14~15+N
14	Status. Failure 0x00 Success 0x01
15:15+N	ASCIIZ Variable Value

1.11 YAFU_Protect_Flash Command

This command is used to enable or disable Flash write protection.

NetFn	0x32
Command	0x25

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0025 (LSB first)
7:8	Datalen. Fixed 0x0005 (LSB first)
9:12	CRC32chksum. Checksum of Byte13~17
13:16	Memory Block Number which needs protection to be turned on/off.
17	Protect. OFF 0x00 ON 0x01

***Note**: To protect all blocks, specify a Block number of 0xFFFFFFFF.*

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0025 (LSB first)
8:9	Datalen. Fixed 0x0001 (LSB first)
10:13	CRC32chksum
14	Status. Success 0x00 Failed 0x01

1.12 YAFU_Allocate_Memory Command

This command is used to allocate memory.

NetFn	0x32
Command	0x20

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0020 (LSB first)
7:8	Datalen. Fixed 0x0004 (LSB first)
9:12	CRC32chksum. Checksum of Byte13~16
13:16	Size of Memory Required. (LSB first)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0020 (LSB first)
8:9	Datalen. Fixed 0x0004 (LSB first)
10:13	CRC32chksum. Checksum of Byte14~17
14:17	Address of Memory Allocated. (LSB first)

Note: *On error, it returns the address as -1 (0xFFFFFFFF).*

1.13 YAFU_Write_Memory Command

This command is used to transfer the image or modules to the allocated memory.

NetFn	0x32
Command	0x31

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0031 (LSB first)
7:8	Datalen. =5+N (N < 64K) LSB first
9:12	CRC32chksum. Checksum of Byte13~18+N
13:16	Memory Offset to write. (LSB first)
17	Write width.
18:18+N	Memory Contents to write

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0031 (LSB first)
8:9	Datalen. Fixed 0x0002 (LSB first)
10:13	CRC32chksum
14:15	Size Written. (LSB first)

1.14 YAFU_Miscellaneous_Info Command

This command is used to set Preserve Flag.

NetFn	0x32
Command	0x59

Request Data:

Byte	Data field
1	Preserve Flag
2:3	Reserved1
4:5	Reserved2
6:7	Reserved3
8:9	Reserved4
10:11	Reserved5
12:13	Reserved6
14:15	Reserved7
16:17	Reserved8

Response Data:

Byte	Data field
1	Completion Code
2:3	Reserved1
4:5	Reserved2
6:7	Reserved3
8:9	Reserved4

1.15 YAFU_Verify_Flash Command

This command is used to verify the new image.

NetFn	0x32
Command	0x27

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0027 (LSB first)
7:8	Datalen. Fixed 0x000c (LSB first)
9:12	CRC32chksum. Checksum of Byte13~24
13:16	Memory Offset. Offset address which contains the data against which verification has to be done (LSB first)
17:20	Flash Offset. Flash memory content offset address which needs to be verified. (LSB first)
21:24	Size to Verify. (LSB first)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0027 (LSB first)
8:9	Datalen. Fixed 0x0004 (LSB first)
10:13	CRC32chksum
14:17	Offset. (LSB first)

Note: *Success if Offset = Size. If Failure, Offset gives the offset where the mismatch happened.*

1.16 YAFU_Get_Verify_Status Command

This command is used to get the verify status to check verify progress completed.

NetFn	0x32
Command	0x29

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0029 (LSB first)
7:8	Datalen. Fixed 0x0000
9:12	CRC32chksum. (byte values will be ignored)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0029 (LSB first)
8:9	Datalen. Fixed 0x0000
10:13	CRC32chksum. Checksum of Byte16~19
14:17	Status. (LSB first) Failure 0x0000 Success 0x0001
18:21	Offset. (LSB first)
22:23	Progress (% completed)

Note: *Success if Offset = Size. If Failure, Offset gives the offset where the mismatch happened.*

1.17 YAFU_Free_Memory Command

This command is used to free up memory.

NetFn	0x32
Command	0x21

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0010 (LSB first)
7:8	Datalen. Fixed 0x0002 (LSB first)
9:12	CRC32chksum. Checksum of Byte13~14
13:16	Mode Type. LSB first 0x0000 Normal mode 0x0001 Recovery mode
15	Migration

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0021 (LSB first)
8:9	Datalen. Fixed 0x0001 (LSB first)
10:13	CRC32chksum
14	Status. Success 0x00 Failed 0x01

1.18 YAFU_Set_Boot_Config Command

This command is used to restore the necessary Boot configuration.

NetFn	0x32
Command	0x41

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0041 (LSB first)
7:8	Datalen. =65+N
9:12	CRC32chksum. Checksum of Byte13~78+N
13:77	ASCIIZ Variable Name
78:78+N	ASCIIZ Variable Value

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum.
6:7	YafuCmd. Fixed 0x0041 (LSB first)
8:9	Datalen. Fixed 0x0001 (LSB first)
10:13	CRC32chksum
14	Status. Failure 0x00 Success 0x01

1.19 YAFU_Deactivate_Flash_Mode Command

This command is used to inform device to leave the Flash mode.

NetFn	0x32
Command	0x50

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0050 (LSB first)
7:8	Datalen. Fixed 0x0000 (LSB first)
9:12	CRC32chksum. (byte values will be ignored)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum.
6:7	YafuCmd. Fixed 0x0050 (LSB first)
8:9	Datalen. Fixed 0x0001 (LSB first)
10:13	CRC32chksum
14	Status. Success 0x00 Failure 0x01

1.20 YAFU_Restore_Flash_Device Command

This command is used to restore Flash device.

NetFn	0x32
Command	0x53

Request Data:

Byte	Data field
1	SPIDevice

Response Data:

Byte	Data field
1	Completion Code.
2:32	MTDName. Represented in ASCII value
33:36	Flash Size. (LSB first)
37:40	Flash Address. Start address of flash (LSB first)
41:44	Flash Erase Block Size. (LSB first)
45:46	Flash Product ID. (LSB first)
47	Flash Width.
48	FMH Compliance.
49:50	Reserved. Fixed 0x00
51:52	Number Of Erase Blocks(N). (LSB first)

1.21 YAFU_Reset_Device Command

This command is used to reset device.

NetFn	0x32
Command	0x51

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0051 (LSB first)
7:8	Datalen. Fixed 0x0002 (LSB first)
9:12	CRC32chksum
13:14	Wait time in Seconds

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0051 (LSB first)
8:9	Datalen. Fixed 0x0001 (LSB first)
10:13	CRC32chksum
14	Status

Note: *Status =0, if the device is busy and cannot be reset. Status =1, the device will be reset after waiting for "Wait time" Seconds.*

1.22 YAFU_Activate_Flash_Device Command

This command is used to Activate Flash for the specific device.

NetFn	0x32
Command	0x56

Request Data:

Byte	Data field
1	SPI Device 0x00: BMC 0x01: BOIS 0x02: CPLD 0x03: BMV_MMC

Response Data:

Byte	Data field
1	Completion Code 0x00: SUCCESS 0x23: YAFU Not Supported 0x03: Device Open error 0x11: Flash or erase failure 0x08: BMC is in Deactivate state

1.23 YAFU_Get_Firmware_Info Command

This command is used to get information about the Firmware.

NetFn	0x32
Command	0x02

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0002 (LSB first)
7:8	Datalen. Fixed 0x0000
9:12	CRC32chksum.

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0002 (LSB first)
8:9	Datalen. Fixed 0x0026 (LSB first)
10:13	CRC32chksum
14	Firmware Major Version
15	Firmware Minor Version
16:17	Firmware Auxiliary (LSB first)
18-21	Firmware BUILD Number (LSB first)
22-25	Reserved
26-33	Firmware Name String
34-37	Firmware Size (LSB first)
38-41	Product ID (LSB first)

1.24 YAFU_Write_Flash Command

This command is used to write Flash Contents.

NetFn	0x32
Command	0x23

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0023 (LSB first)
7:8	Datalen
9:12	CRC32chksum. Checksum of Byte13~21
13:16	Offset to Write. Memory offset where the data needs to be written (LSB first)
17	Write Width

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0023 (LSB first)
8:9	Datalen. Fixed 0x0002 (LSB first)
10:13	CRC32chksum. Checksum of Byte14:15
14:15	Number of bytes written (LSB first)

1.25 YAFU_Erase_Flash Command

This command is used to erase Flash Contents.

NetFn	0x32
Command	0x24

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0024 (LSB first)
7:8	Datalen. Fixed 0x0004
9:12	CRC32chksum
13:16	Block number to erase (LSB first)

Note: *To erase all blocks, specify block number as 0xFFFFFFFF.*

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0024 (LSB first)
8:9	Datalen. Fixed 0x0001(LSB first)
10:13	CRC32chksum. Checksum of Byte14
14	Status 0x00: Failed 0x01: Success

1.26 YAFU_Read_Memory Command

This command is used to read the image memory.

NetFn	0x32
Command	0x30

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0030 (LSB first)
7:8	Datalen. Fixed 0x0007 (LSB first)
9:12	CRC32chksum. Checksum of Byte13~19
13:16	Memory Offset to read. (LSB first)
17	Read width
18:19	Size to read. (LSB first)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0030 (LSB first)
8:9	Datalen. Size to read
10:13	CRC32chksum
14:N	Memory contents read

1.27 YAFU_Copy_Memory Command

This command is used to copy image memory from one location to other.

NetFn	0x32
Command	0x32

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0032 (LSB first)
7:8	Datalen. Fixed 0x000c (LSB first)
9:12	CRC32chksum. Checksum of Byte13~24
13:16	Memory Offset for source (LSB first)
17:20	Memory Offset for destination (LSB first)
21:24	Size to copy (LSB first)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0032 (LSB first)
8:9	Datalen. Fixed 0x0004 (LSB first)
10:13	CRC32chksum
14:17	Size copied (LSB first)

1.28 YAFU_Compare_Memory Command

This command is used to compare memory offsets are holding same data or not.

NetFn	0x32
Command	0x33

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0033 (LSB first)
7:8	Datalen. Fixed 0x000C (LSB first)
9:12	CRC32chksum. Checksum of Byte13~24
13:16	Memory Offset for First Offset (LSB first)
17:20	Memory Offset for Second Offset (LSB first)
21:24	Size to compare (LSB first)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum.
6:7	YafuCmd. Fixed 0x0033 (LSB first)
8:9	Datalen. Fixed 0x0004 (LSB first)
10:13	CRC32chksum
14:17	Offset Value (LSB first)

Note: *Success if Offset = Size. If Failure, Offset gives the offset where the mismatch happened.*

1.29 YAFU_Clear_Memory Command

This command is used to clear up memory.

NetFn	0x32
Command	0x34

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0034 (LSB first)
7:8	Datalen. Fixed 0x0008 (LSB first)
9:12	CRC32chksum. Checksum of Byte13~20
13:16	Address Offset from where to be cleared. (LSB first)
17:20	Size, how much to be cleared. (LSB first)

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0034 (LSB first)
8:9	Datalen. Fixed 0x0004(LSB first)
10:13	CRC32chksum
14:17	Size, how much got cleared. (LSB first)

1.30 YAFU_Dual_Img_Support Command

This command is used for dual image support.

NetFn	0x32
Command	0x54

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0054(LSB first)
7:8	Datalen. Fixed 0x04
9:12	CRC32chksum. Checksum of Byte13~16
13	PreserveConf 0x00:FALSE 0x01:TRUE
14:16	Reserved

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0054 (LSB first)
8:9	Datalen. Fixed 0x01
10:13	CRC32chksum

1.31 YAFU_Firmware_Select_Flash Command

This command is used to select flash memory.

NetFn	0x32
Command	0x55

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0055 (LSB first)
7:8	Datalen. Fixed 0x0001 (LSB first)
9:12	CRC32chksum
13	Firmware Select

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x0055 (LSB first)
8:9	Datalen. Fixed 0x0001 (LSB first)
10:13	CRC32chksum

1.32 YAFU_Compare_ME_Version Command

This command is used to Compare ME Version.

NetFn	0x32
Command	0x35

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x0035 (LSB first)
7:8	Datalen. Fixed 0x0008 (LSB first)
9:12	CRC32chksum. Checksum of Byte13~16
13:16	Address Offset from where to compare
17:20	Size of Image

1.33 YAFU_Get_Image_Size Command

This command is used to send image size and interface channel number to BMC.

NetFn	0x32
Command	0x1E

Request Data:

Byte	Data field
1	Image size
2	Channel Number

Response Data:

Byte	Data field
1	Completion Code

1.34 YAFU_Start_Stateless Command

This command is used to start stateless YAFU flash.

NetFn	0x32
Command	0x2F

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x002C (LSB first)
7:8	Datalen.
9:12	CRC32chksum (byte values will be ignored)
13	preserveConf. Flag to confirm configuration is preserved or not
14	RebootConf. Flag to Confirm BMC is rebooted or no

Response Data:

Byte	Data field
1	Completion Code
2	Reserved

1.35 YAFU_Write_File Command

This command is used to write the data in to the file.

NetFn	0x32
Command	0x2D

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x002D (LSB first)
7:8	Datalen.
9:12	CRC32chksum Checksum of Byte13~21
13:16	Offset to Write. File offset where the data needs to be written in file (LSB first)
17	Write Width

Response Data:

Byte	Data field
1	Completion Code
2:5	Requested Seqnum
6:7	YafuCmd. Fixed 0x002D (LSB first)
8:9	Datalen. Fixed 0x0002 (LSB first)
10:13	CRC32chksum. Checksum of Byte14:15
14:15	Number of bytes written (LSB first)

1.36 YAFU_Get_Flsh_Status Command

This command is used to current state status used in stateless flash.

NetFn	0x32
Command	0x2E

Request Data:

Byte	Data field
1:4	Seqnum. Cmd Sequence Number
5:6	YafuCmd. Fixed 0x002E (LSB first)
7:8	Datalen. Fixed 0x0000
9:12	CRC32chksum.(byte values will be ignored)

Response Data:

Byte	Data field
1	Completion Code
2	Action (Idle = 0, Preparing = 1, Downloading = 2, Verifying = 3, Flashing = 4, Flash Verifying = 5)
3	progress (Pending = 1, Ok = 2, Failure = 3)
4	Overall
5:64	SubAction
65:94	Progress (Flash state information)
95	State (Doing =0, ToBeDone = 1, Complete = 2, Skipped = 3, Error = 4)
96	Reserved

1.37 YAFU_Check_Map_Difference command

This command is used to set the SPIMAPDifference flag when conf/extlog partition changed.

NetFn	0x32
Command	0x05

Request Data:

Byte	Data field
1	SPIMAPDifference flag to inform conf/extlog partition changed or not

Response Data:

Byte	Data field
1	Completion Code

1.38 CMD_AMI_GET_COMP_DEVLIST

To get the device list of components such as BIOS, PLDM, CPLD or NVDIMM, Supported below IPMI command.

int AMIGetCompDevList(_NEAR_ INT8U *pReq, INT32U ReqLen, _NEAR_ INT8U *pRes,_NEAR_ int BMCInst)

Command:

NetFn	0x32
Command	0x7

Request Data :

Byte	Data Field
1	Component Type

Response Data

Byte	Data Field
1	Completion code 0x00 – Success

	0x81 – No Device Found 0x82 – Invalid Component 0x83 – Operation Failed
2	Component Type
3	Device Count
4-N	4 - Firmware Update Progress Status (Device1) 5:20 - Device ID (Maximum 10 DeviceId)

1.39 CMD_AMI_INITIATE_PARALLEL_FW_OPERATION

Set component type, device id, total image size and flash offset to do firmware update.

int AMIStartParallelFirmwareOperation(_NEAR_ INT8U *pReq, INT32U ReqLen, _NEAR_ INT8U *pRes,_NEAR_ int BMCInst)

Command:

NetFn	0x32
Command	0x8

Request Data

Byte	Data Field
1-2	Operation
3-4	ComponentType
5-44	Sizetocopy (Every 4 bytes maps one device in sequence)
45-48	Flashoffset
49	DevCount
50-209	50:65 – DeviceId[0] (Every 16 bytes maps one device in sequence, maximum 10 DeviceId)

Response Data

Byte	Data Field
1	Completion code

	0x00 – Success 0x81 – No Device Found 0x82 – Invalid Component 0x83 – Operation Failed

1.40 CMD_AMI_GET_PARALLEL_OPERATION_STATUS

Get the upload image and firmware update progress.

int AMIGetParallelOperationStatus(_NEAR_ INT8U *pReq, INT32U ReqLen, _NEAR_ INT8U *pRes,_NEAR_ int BMCInst)

Command:

NetFn	0x32
Command	0x9

Request Data

Byte	Data Field
1-2	DevCount
3SS	DeviceId

Response Data

Byte		Data Field
1		Completion code 0x00 – Success 0x81 – No Device Found 0x82 – Invalid Component 0x83 – Operation Failed
2-3		DevCount
4-N	4:56 (Data structure of the first device)	Data 4:5 - action Data 5:6 - state Data 7:8 - progress Data 9:40 - subaction Data 41:56 - DeviceId

	(Maximum 10 device)

2 YAFU Error Codes

Error Codes	Macro Used	Definition
0x00	-	On Success/Normal Response
0x01	YAFU_FW_MOD_NOT_FOUND	Firmware Module Not Found
0x02	YAFU_GREATER_IMAGE_SIZE	Image Size is Greater
0x03	YAFU_GET_DUAL_IMAGE_FAILED	Dual Image Configurations failed
0x04	YAFU_IMAGE_CHKSUM_VERIFY_ FAILED	Image Checksum verification failed
0x05	YAFU_FILE_OPEN_ERR	Cannot Open File
0x06	YAFU_INVALID_NAME	Invalid Name Given Invalid Host Name Invalid Publickey File Name Invalid IP Address
0x07	YAFU_NAME_LONG	Parameter Size Exceeds Public key File Name Size Exceeds IP Address Size Exceeds Host Name Size Exceeds Username Size Exceeds Password Size Exceeds
0x08	YAFU_CC_IMAGE_SIZE_INVALID	Invalid Image Size
0x09	YAFU_COMMAND_TIMEOUT_ERR	Command Timeout Exception
0xA	YAFU_CC_IMAGE_SIGNED	Image is signed but signed image support not enabled
0xB	YAFU_CC_NOT_SIGNED_IMAGE	Image is not signed but signed image support is enabled
0x10	YAFU_INVALID_IP	Invalid IP is provided
0xE	YAFU_IMAGE_SIZE_SENDING_FAILED	Sending image size and/or channel num failed

3 YAFU Command Specific Error Codes

Command	Macro used	Error Codes	Error
CMD_AMI_YAFU_GET_STATUS	YAFU_GET_STATUS_IPMICMD_ERROR	0x100	IPMI Command Error
	YAFU_GET_STATUS_TIMEOUT_ERROR	0x101	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_GET_STATUS_NAK_ERROR	0x102	YAFU Not Acknowledge Error
	YAFU_GET_STATUS_CHKSUM_ERROR	0x103	Checksum Error
	YAFU_GET_STATUS_FLASHER_NOT_READY_ERROR	0x105	YAFU_CC_FLASHER_NOT_READY
	YAFU_GET_STATUS_CHANNEL_DISABLED	0x106	IPMI Messaging Disabled for current channel
CMD_AMI_GET_FEATURE_STATUS	AMI_GET_FEATURE_STATUS_IPMICMD_ERROR	0x110	IPMI Command Error
	AMI_GET_FEATURE_STATUS_TIMEOUT_ERROR	0x111	LIBIPMI_MEDIUM_E_TIMED_OUT
	AMI_GET_FEATURE_STATUS_FWUPDATE_MODE_ERROR	0x114	YAFU_CC_DEV_IN_FIRMWARE_UPDATE_MODE
	AMI_GET_FEATURE_STATUS_KCSSMM_CH_ERROR	0x116	YAFU Commands are not supported via KCS SMM Channel
CMD_AMI_YAFU_SWITCH_FLASH_DEVICE	YAFU_SWITCH_FLASH_DEVICE_IPMICMD_ERROR	0x120	IPMI Command Error
	YAFU_SWITCH_FLASH_DEVICE_TIMEOUT_ERROR	0x121	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_SWITCH_FLASH_DEVICE_FWUPDATE_MODE_ERROR	0x124	YAFU_CC_DEV_IN_FIRMWARE_UPDATE_MODE
	YAFU_SWITCH_FLASH_DEVICE_FLASHER_NOT_READY_ERROR	0x125	YAFU_CC_FLASHER_NOT_READY
CMD_AMI_DUAL_IMG_SUPPORT	AMI_DUAL_IMG_SUPPORT_IPMICMD_ERROR	0x130	IPMI Command Error

	AMI_DUAL_IMG_SUPPORT_TIMEOUT_ERROR	0x131	LIBIPMI_MEDIUM_E_TIMED_OUT
CMD_AMI_YAFU_DUAL_IMAGE_SUP	YAFU_DUAL_IMAGE_SUP_IPMICMD_ERROR	0x140	IPMI Command Error
	YAFU_DUAL_IMAGE_SUP_TIMEOUT_ERROR	0x141	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_DUAL_IMAGE_SUP_NAK_ERROR	0x142	YAFU Not Acknowledge Error
CMD_AMI_YAFU_GET_FLASH_INFO	YAFU_GET_FLASH_INFO_IPMICMD_ERROR	0x150	IPMI Command Error
	YAFU_GET_FLASH_INFO_NAK_ERROR	0x152	YAFU Not Acknowledge Error
	YAFU_GET_FLASH_INFO_CHKSUM_ERROR	0x153	Checksum Error
CMD_AMI_YAFU_GET_FMH_INFO	YAFU_GET_FMH_INFO_IPMICMD_ERROR	0x160	IPMI Command Error
	YAFU_GET_FMH_INFO_NAK_ERROR	0x162	YAFU Not Acknowledge Error
	YAFU_GET_FMH_INFO_CHKSUM_ERROR	0x163	Checksum Error
CMD_AMI_YAFU_ACTIVATE_FLASH	YAFU_ACTIVATE_FLASH_IPMICMD_ERROR	0x170	IPMI Command Error
	YAFU_ACTIVATE_FLASH_TIMEOUT_ERROR	0x171	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_ACTIVATE_FLASH_NAK_ERROR	0x172	YAFU Not Acknowledge Error
	YAFU_ACTIVATE_FLASH_FWUPDATE_MODE_ERROR	0x174	YAFU_CC_DEV_IN_FIRMWARE_UPDATE_MODE
	YAFU_ACTIVATE_FLASH_FLASHER_NOT_READY_ERROR	0x175	YAFU_CC_FLASHER_NOT_READY
	YAFU_ACTIVATE_FLASH_ERROR_CASE_6	0x176	pAMIYAFUActivateFlashDeviceRes.CompletionCode == 0x08
	YAFU_ACTIVATE_FLASH_ERROR_CASE_7	0x177	pAMIYAFUActivateFlashDeviceRes.CompletionCode == 0x03
	YAFU_ACTIVATE_FLASH_ERROR_CASE_8	0x178	pAMIYAFUActivateFlashDeviceRes.CompletionCode == 0x23

CMD_AMI_YAFU_GET_BOOT_CONFIG	YAFU_GET_BOOT_CONFIG_IPMICMD_ERROR	0x180	IPMI Command Error
	YAFU_GET_BOOT_CONFIG_NAK_ERROR	0x182	YAFU Not Acknowledge Error
	YAFU_GET_BOOT_CONFIG_CHKSUM_ERROR	0x183	Checksum Error
CMD_AMI_YAFU_SET_BOOT_CONFIG	YAFU_SET_BOOT_CONFIG_IPMICMD_ERROR	0x190	IPMI Command Error
	YAFU_SET_BOOT_CONFIG_TIMEOUT_ERROR	0x191	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_SET_BOOT_CONFIG_NAK_ERROR	0x192	YAFU Not Acknowledge Error
	YAFU_SET_BOOT_CONFIG_CHKSUM_ERROR	0x193	Checksum Error
CMD_AMI_YAFU_GET_BOOT_VARS	YAFU_GET_BOOT_VARS_IPMICMD_ERROR	0x1A0	IPMI Command Error
	YAFU_GET_BOOT_VARS_TIMEOUT_ERROR	0x1A1	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_GET_BOOT_VARS_NAK_ERROR	0x1A2	YAFU Not Acknowledge Error
	YAFU_GET_BOOT_VARS_CHKSUM_ERROR	0x1A3	Checksum Error
CMD_AMI_YAFU_PROTECT_FLASH	YAFU_PROTECT_FLASH_IPMICMD_ERROR	0x1B0	IPMI Command Error
	YAFU_PROTECT_FLASH_TIMEOUT_ERROR	0x1B1	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_PROTECT_FLASH_NAK_ERROR	0x1B2	YAFU Not Acknowledge Error
	YAFU_PROTECT_FLASH_CHKSUM_ERROR	0x1B3	Checksum Error
CMD_AMI_YAFU_ALLOCATE_MEMORY	YAFU_ALLOCATE_MEMORY_IPMICMD_ERROR	0x1C0	IPMI Command Error
	YAFU_ALLOCATE_MEMORY_NAK_ERROR	0x1C2	YAFU Not Acknowledge Error

	YAFU_ALLOCATE_MEMORY_CHKSUM_ERROR	0x1C3	Checksum Error
CMD_AMI_YAFU_WRITE_MEMORY	YAFU_WRITE_MEMORY_IPMICMD_ERROR	0x1D0	IPMI Command Error
	YAFU_WRITE_MEMORY_NAK_ERROR	0x1D2	YAFU Not Acknowledge Error
	YAFU_WRITE_MEMORY_CHKSUM_ERROR	0x1D3	Checksum Error
CMD_AMI_YAFU_MISCELLANEOUS_INFO	YAFU_MISCELLANEOUS_INFO_IPMICMD_ERROR	0x1E0	IPMI Command Error
	YAFU_MISCELLANEOUS_INFO_TIMEOUT_ERROR	0x1E1	LIBIPMI_MEDIUM_E_TIMED_OUT
CMD_AMI_YAFU_ERASE_COPY_FLASH	YAFU_ERASE_COPY_FLASH_IPMICMD_ERROR	0x1F0	IPMI Command Error
	YAFU_ERASE_COPY_FLASH_TIMEOUT_ERROR	0x1F1	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_ERASE_COPY_FLASH_NAK_ERROR	0x1F2	YAFU Not Acknowledge Error
	YAFU_ERASE_COPY_FLASH_CHKSUM_ERROR	0x1F3	Checksum Error
	YAFU_ERASE_COPY_FLASH_INVALID_SIGN_IMAGE_ERROR	0x1F6	YAFU_CC_INVALID_SIGN_IMAGE
	YAFU_ERASE_COPY_FLASH_SIGNED_SUPP_NOT_ENABLED_ERROR	0x1F7	YAFU_CC_SIGNED_SUPP_NOT_ENABLED
	YAFU_ERASE_COPY_FLASH_RAID_CTRL_ID_INVALID_ERROR	0x1F8	RAID_CTRL_ID_INVALID
CMD_AMI_YAFU_GET_ECF_STATUS	YAFU_GET_ECF_STATUS_IPMICMD_ERROR	0x200	IPMI Command Error
	YAFU_GET_ECF_STATUS_TIMEOUT_ERROR	0x201	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_GET_ECF_STATUS_NAK_ERROR	0x202	YAFU Not Acknowledge Error
	YAFU_GET_ECF_STATUS_CHKSUM_ERROR	0x203	Checksum Error

	YAFU_GET_ECF_STATUS_FLASH_SAME_IMAGE	0x206	YAFU_CC_FLASH_SAME_IMAGE
CMD_AMI_YAFU_VERIFY_FLASH	YAFU_VERIFY_FLASH_IPMICMD_ERROR	0x210	IPMI Command Error
	YAFU_VERIFY_FLASH_TIMEOUT_ERROR	0x211	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_VERIFY_FLASH_NAK_ERROR	0x212	YAFU Not Acknowledge Error
	YAFU_VERIFY_FLASH_CHKSUM_ERROR	0x213	Checksum Error
CMD_AMI_YAFU_GET_VERIFY_STATUS	YAFU_GET_VERIFY_STATUS_IPMICMD_ERROR	0x220	IPMI Command Error
	YAFU_GET_VERIFY_STATUS_TIMEOUT_ERROR	0x221	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_GET_VERIFY_STATUS_NAK_ERROR	0x222	YAFU Not Acknowledge Error
	YAFU_GET_VERIFY_STATUS_CHKSUM_ERROR	0x223	Checksum Error
	YAFU_GET_VERIFY_STATUS_KNOWN_ERROR	0x226	Exiting the tool for Known Error Status Codes
CMD_AMI_YAFU_FREE_MEMORY	YAFU_FREE_MEMORY_IPMICMD_ERROR	0x230	IPMI Command Error
	YAFU_FREE_MEMORY_NAK_ERROR	0x232	YAFU Not Acknowledge Error
	YAFU_FREE_MEMORY_CHKSUM_ERROR	0x233	Checksum Error
CMD_AMI_YAFU_DEACTIVATE_FLASH_MODE	YAFU_DEACTIVATE_FLASH_MODE_IPMICMD_ERROR	0x240	IPMI Command Error
	YAFU_DEACTIVATE_FLASH_MODE_TIMEOUT_ERROR	0x241	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_DEACTIVATE_FLASH_MODE_NAK_ERROR	0x242	YAFU Not Acknowledge Error
	YAFU_DEACTIVATE_FLASH_MODE_CHKSUM_ERROR	0x243	Checksum Error

	YAFU_DEACTIVATE_FLASH_MODE_RES_LEN_ERROR	0x246	ResLen != sizeof(AMIYAFUDeactivateFlashRes_T) - 1
CMD_AMI_YAFU_RESTORE_FLASH_DEVICE	YAFU_RESTORE_FLASH_DEVICE_IPMICMD_ERROR	0x250	IPMI Command Error
CMD_AMI_YAFU_RESET_DEVICE	YAFU_RESET_DEVICE_IPMICMD_ERROR	0x260	IPMI Command Error
	YAFU_RESET_DEVICE_NAK_ERROR	0x262	YAFU Not Acknowledge Error
	YAFU_RESET_DEVICE_CHKSUM_ERROR	0x263	Checksum Error
	YAFU_RESET_DEVICE_FWUPDATE_MODE_ERROR	0x264	YAFU_CC_DEV_IN_FIRMWARE_UPDATE_MODE
	YAFU_RESET_DEVICE_FLASHER_NOT_READY_ERROR	0x265	YAFU_CC_FLASHER_NOT_READY
	YAFU_RESET_DEVICE_RES_LEN_ERROR	0x266	ResLen != sizeof(AMIYAFUResetDeviceRes_T)
CMD_AMI_GET_FW_VERSION	AMI_GET_FW_VERSION_IPMICMD_ERROR	0x270	IPMI Command Error
CMD_AMI_YAFU_WRITE_FILE	YAFU_WRITE_FILE_IPMICMD_ERROR	0x280	IPMI Command Error
	YAFU_WRITE_FILE_NAK_ERROR	0x282	YAFU Not Acknowledge Error
	YAFU_WRITE_FILE_CHKSUM_ERROR	0x283	Checksum Error
	YAFU_WRITE_FILE_FWUPDATE_MODE_ERROR	0x284	YAFU_CC_DEV_IN_FIRMWARE_UPDATE_MODE
CMD_AMI_YAFU_GET_FLSH_STATUS	YAFU_GET_FLSH_STATUS_IPMICMD_ERROR	0x290	IPMI Command Error
	YAFU_GET_FLSH_STATUS_TIMEOUT_ERROR	0x291	LIBIPMI_MEDIUM_E_TIMED_OUT
	YAFU_GET_FLSH_STATUS_FWUPDATE_MODE_ERROR	0x294	YAFU_CC_DEV_IN_FIRMWARE_UPDATE_MODE
	YAFU_GET_FLSH_STATUS_FLASH_ERROR	0x296	pAMIYAFUGetFlshStatusRes.State == FLASH_ERROR

CMD_AMI_YAFU_START_STATELESS	YAFU_START_STATELESS_IPMICMD_ERROR	0x2A0	IPMI Command Error